

Design Data for CAD model and drawing preparation

Human-readable handoff of the saved R1 geometry. Use the dimensions and coordinate schedules, not scaled illustrations. This report does not regenerate, resize or qualify the equipment.

Document authority	Saved record
Design / revision / record	FIXTURE-47 / 3 / fixture
Saved on	2026-09-19T12:00:00.000Z
Currentness	VERIFICATION FIXTURE — NOT A LIVE DESIGN
Geometry status	SYSTEM-GENERATED PRE-PILOT GEOMETRY — GEOMETRICALLY COMPLETE
Geometry ruleset	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1
Source Stage 3 ID	3
Source Stage 4 ID	4
Combined source SHA-256	aa
Geometry SHA-256	a164f386580c29d7cae57285064329a0dcda0e239c5634e9a6f71c198187d42e
Rules manifest SHA-256	86398f4b40af5d3dac8cee984486837d59799cb988b1b6664bd424068b98d93d

A = frozen upstream authority; B = documented reference feature; C = approved R1 engineering envelope. A calculated Class-C value remains Class C. Geometry completeness does not establish hydraulic, structural or fabrication adequacy.

VERIFICATION FIXTURE — NOT A LIVE DESIGN

Coordinate and datum convention

Right-handed Cartesian coordinates. Origin (0,0,0) is the bottom vessel pole on the shaft/column axis. +Z is vertically upward; X/Y are the horizontal plane. Azimuth 0° is +X, 90° is +Y, increasing counter-clockwise when viewed from above (+Z toward origin).

All length tables and XYZ coordinates below are millimetres; angular values are degrees. The immutable source stores SI metres. Negative Z values belong to the skirt below the bottom vessel pole. Radial shell connections point outward; “up” and “down” connections have vertical axes. Connection centre denotes saved vessel-surface intersection, and end denotes saved external stub termination, not a flange face.

Numeric presentation is rounded to at most six decimal places, without modifying source values or adding manufacturing tolerances. Symbols in the provenance rules refer to SI source quantities: D = column ID, DR = rotor swept diameter, hc = compartment pitch, ds = shaft diameter, do = opening diameter, ts = stator thickness, HR = rotor axial envelope, dh = hub diameter, tb = blade thickness, Lb = blade radial length, HA = installed active height. Do not mix mm table values with SI rule quantities.

Rotor vertex XYZ values are local to the rotor centre plane (Z=0) on the shared shaft axis. Translate local Z by the rotor elevation in the complete schedule; do not apply additional clocking. Support footprints are global XY; combine them with the saved arm bottom/top Z values. Head profiles are internal envelopes, not material outer surfaces.

Design basis and source provenance

Design feature	Saved value	Unit	Provenance / governing rule
Stage3 Result Id	3	—	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Stage4 Result Id	4	—	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Column Diameter	600	mm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Rotor Diameter	300	mm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Rotor Diameter Ratio	0.5	—	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Compartment Height	180	mm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Compartment Count	39	—	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Required Active Height	7000	mm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Installed Active Height	7020	mm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Design Nt	7	—	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Hets	1000	mm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Selected Rpm	50	rpm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Rpm Min	30	rpm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Rpm Max	70	rpm	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Phase Configuration	NMP continuous / RRBO dispersed	—	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Column Area area	282743.338823	mm ²	Class C; System-generated Class C. $\pi D^2/4$.
Compartment1-bottom	750	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment1-top	930	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment1-rotor	840	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment2-bottom	930	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment2-top	1110	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment2-rotor	1020	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment3-bottom	1110	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment3-top	1290	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment3-rotor	1200	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment4-bottom	1290	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment4-top	1470	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment4-rotor	1380	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment5-bottom	1470	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment5-top	1650	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment5-rotor	1560	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment6-bottom	1650	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment6-top	1830	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment6-rotor	1740	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment7-bottom	1830	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment7-top	2010	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment7-rotor	1920	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment8-bottom	2010	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment8-top	2190	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment8-rotor	2100	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment9-bottom	2190	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment9-top	2370	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment9-rotor	2280	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment10-bottom	2370	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment10-top	2550	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment10-rotor	2460	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment11-bottom	2550	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment11-top	2730	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment11-rotor	2640	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment12-bottom	2730	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment12-top	2910	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment12-rotor	2820	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment13-bottom	2910	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment13-top	3090	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment13-rotor	3000	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment14-bottom	3090	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment14-top	3270	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment14-rotor	3180	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment15-bottom	3270	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment15-top	3450	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment15-rotor	3360	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment16-bottom	3450	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment16-top	3630	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment16-rotor	3540	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment17-bottom	3630	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment17-top	3810	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment17-rotor	3720	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment18-bottom	3810	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment18-top	3990	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment18-rotor	3900	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment19-bottom	3990	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment19-top	4170	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment19-rotor	4080	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment20-bottom	4170	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment20-top	4350	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment20-rotor	4260	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment21-bottom	4350	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment21-top	4530	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment21-rotor	4440	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment22-bottom	4530	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment22-top	4710	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment22-rotor	4620	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment23-bottom	4710	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment23-top	4890	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment23-rotor	4800	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment24-bottom	4890	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment24-top	5070	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment24-rotor	4980	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment25-bottom	5070	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment25-top	5250	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment25-rotor	5160	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment26-bottom	5250	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment26-top	5430	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment26-rotor	5340	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment27-bottom	5430	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment27-top	5610	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment27-rotor	5520	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment28-bottom	5610	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment28-top	5790	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment28-rotor	5700	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment29-bottom	5790	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment29-top	5970	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment29-rotor	5880	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment30-bottom	5970	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment30-top	6150	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment30-rotor	6060	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment31-bottom	6150	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment31-top	6330	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment31-rotor	6240	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment32-bottom	6330	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment32-top	6510	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment32-rotor	6420	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment33-bottom	6510	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment33-top	6690	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment33-rotor	6600	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment34-bottom	6690	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment34-top	6870	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment34-rotor	6780	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment35-bottom	6870	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment35-top	7050	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment35-rotor	6960	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment36-bottom	7050	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment36-top	7230	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment36-rotor	7140	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment37-bottom	7230	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment37-top	7410	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment37-rotor	7320	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment38-bottom	7410	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment38-top	7590	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment38-rotor	7500	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment39-bottom	7590	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment39-top	7770	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment39-rotor	7680	mm	Class C; System-generated Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Vessel, head and active/end-zone geometry

Design feature	Saved value	Unit	Provenance / governing rule
Bottom Disengagement	600	mm	Class C; System-generated Class C. D; packaging allocation, not separation-time qualification.
Top Disengagement	600	mm	Class C; System-generated Class C. D; packaging allocation, not separation-time qualification.
Bottom Head Depth	150	mm	Class C; System-generated Class C. D/4; 2:1 ellipsoidal internal envelope.
Top Head Depth	150	mm	Class C; System-generated Class C. D/4; 2:1 ellipsoidal internal envelope.
Head Straight Flange	0	mm	Class C; System-generated Class C. Zero in R1 internal envelope model.
Active Start	750	mm	Class C; System-generated Class C. hB+LB.
Active End	7770	mm	Class C; System-generated Class C. z0+HA.
Top Tangent	8370	mm	Class C; System-generated Class C. z1+LT.
Vessel Height	8520	mm	Class C; System-generated Class C. HA+2.5D.
Overall Height	9420	mm	Class C; System-generated Class C. Support+vessel+drive = HA+4D.
Head Profiles	2:1 ellipsoidal	—	Class C; System-generated Class C. Internal envelope only.

Head	Tangent Z mm	Pole Z mm	Radial semiaxis mm	Axial semiaxis mm
bottom	150	0	300	150
top	8370	8520	300	150

Each head is the saved half-ellipsoid of revolution: $r^2/a^2 + (z - \text{tangent Z})^2/b^2 = 1$, with semiaxes a/b listed above. Select the half between tangent and pole. This describes the internal geometric envelope only.

The vessel straight internal cylinder uses the frozen column ID and extends between the bottom and top head tangent elevations. No shell outside diameter or material wall thickness is inferred. Model reserved envelopes separately from material-bearing components.

Rotor geometry — six-blade stepped assembly

R1 six-blade stepped rotor — Class C reference-inspired approximation. Actual saved stepped profiles follow; they are R1 engineering approximations, not a claim of exact vendor construction.

Design feature	Saved value	Unit	Provenance / governing rule
Blade Count	6	—	Class B; System-generated Class B. Garthe reference construction family, Table A.1.
Blade Height	46.666667	mm	Class C; System-generated Class C. $\min((7/45) DR, 0.60(hc-ts))$; reference-inspired clearance-capped envelope.
Hub Diameter	106.666667	mm	Class C; System-generated Class C. 1.60 ds; attachment envelope.
Hub Height	37.333333	mm	Class C; System-generated Class C. 0.80 HR; within rotor envelope.
Blade Thickness	3.333333	mm	Class C; System-generated Class C. $\min(DR/90, HR/10, dh/12)$; envelope only.
Blade Radial Length	96.657407	mm	Class C; System-generated Class C. $(\sqrt{DR^2-tb^2}-dh)/2$; exact outer-corner swept diameter.
Blade Inner Segment Length	48.328704	mm	Class C; System-generated Class C. $Lb/2$; approved R1 stepped approximation, not exact reference reproduction.
Blade Outer Segment Length	48.328704	mm	Class C; System-generated Class C. $Lb/2$; approved R1 stepped approximation.
Blade Inner Segment Height	23.333333	mm	Class C; System-generated Class C. $HR/2$, centred about rotor midplane.
Blade Outer Segment Height	46.666667	mm	Class C; System-generated Class C. HR , centred about rotor midplane.
Rotor Offset	90	mm	Class C; System-generated Class C. $hc/2$; centred rotor.
Rotor Axial Envelope	46.666667	mm	Class C; System-generated Class C. Full outer blade axial height.
Rotor Thickness	Not applicable — stepped rotor, not flat disc	—	Class C; System-generated Class C. Explicit N/A, no missing construction dimension.
Rotor Construction	R1 six-blade stepped rotor	—	Class C; System-generated Class C. Reference-inspired family; explicitly approved engineering approximation.
Blade Azimuths angle	0, 60, 120, 180, 240, 300	deg	Class C; System-generated Class C. Six equally spaced blades; same clocking in every compartment.

Blade number	Saved azimuth °
1	0
2	60
3	120
4	180
5	240

Blade number	Saved azimuth °
6	300

Profile vertex	Radial X mm	Local Z mm
1	53.333333	-11.666667
2	101.662037	-11.666667
3	101.662037	-23.333333
4	149.99074	-23.333333
5	149.99074	23.333333
6	101.662037	23.333333
7	101.662037	11.666667
8	53.333333	11.666667

Close the ordered polygon back to vertex 1. The following already-clocked saved 3D vertices define each blade envelope; first and second polygon loops are the opposing thickness faces. Connect corresponding loop vertices. No new blade dimensions are inferred.

Blade / azimuth	Vertex	Local X, Y, Z mm
1 / 0°	1	53.333333, -1.666667, -11.666667
1 / 0°	2	101.662037, -1.666667, -11.666667
1 / 0°	3	101.662037, -1.666667, -23.333333
1 / 0°	4	149.99074, -1.666667, -23.333333
1 / 0°	5	149.99074, -1.666667, 23.333333
1 / 0°	6	101.662037, -1.666667, 23.333333
1 / 0°	7	101.662037, -1.666667, 11.666667
1 / 0°	8	53.333333, -1.666667, 11.666667
1 / 0°	9	53.333333, 1.666667, -11.666667
1 / 0°	10	101.662037, 1.666667, -11.666667
1 / 0°	11	101.662037, 1.666667, -23.333333
1 / 0°	12	149.99074, 1.666667, -23.333333
1 / 0°	13	149.99074, 1.666667, 23.333333
1 / 0°	14	101.662037, 1.666667, 23.333333
1 / 0°	15	101.662037, 1.666667, 11.666667
1 / 0°	16	53.333333, 1.666667, 11.666667
2 / 60°	1	28.110042, 45.354688, -11.666667
2 / 60°	2	52.274394, 87.208573, -11.666667
2 / 60°	3	52.274394, 87.208573, -23.333333
2 / 60°	4	76.438746, 129.062458, -23.333333
2 / 60°	5	76.438746, 129.062458, 23.333333

Blade / azimuth	Vertex	Local X, Y, Z mm
2 / 60°	6	52.274394, 87.208573, 23.333333
2 / 60°	7	52.274394, 87.208573, 11.666667
2 / 60°	8	28.110042, 45.354688, 11.666667
2 / 60°	9	25.223291, 47.021355, -11.666667
2 / 60°	10	49.387643, 88.87524, -11.666667
2 / 60°	11	49.387643, 88.87524, -23.333333
2 / 60°	12	73.551995, 130.729125, -23.333333
2 / 60°	13	73.551995, 130.729125, 23.333333
2 / 60°	14	49.387643, 88.87524, 23.333333
2 / 60°	15	49.387643, 88.87524, 11.666667
2 / 60°	16	25.223291, 47.021355, 11.666667
3 / 120°	1	-25.223291, 47.021355, -11.666667
3 / 120°	2	-49.387643, 88.87524, -11.666667
3 / 120°	3	-49.387643, 88.87524, -23.333333
3 / 120°	4	-73.551995, 130.729125, -23.333333
3 / 120°	5	-73.551995, 130.729125, 23.333333
3 / 120°	6	-49.387643, 88.87524, 23.333333
3 / 120°	7	-49.387643, 88.87524, 11.666667
3 / 120°	8	-25.223291, 47.021355, 11.666667
3 / 120°	9	-28.110042, 45.354688, -11.666667
3 / 120°	10	-52.274394, 87.208573, -11.666667
3 / 120°	11	-52.274394, 87.208573, -23.333333
3 / 120°	12	-76.438746, 129.062458, -23.333333
3 / 120°	13	-76.438746, 129.062458, 23.333333
3 / 120°	14	-52.274394, 87.208573, 23.333333
3 / 120°	15	-52.274394, 87.208573, 11.666667
3 / 120°	16	-28.110042, 45.354688, 11.666667
4 / 180°	1	-53.333333, 1.666667, -11.666667
4 / 180°	2	-101.662037, 1.666667, -11.666667
4 / 180°	3	-101.662037, 1.666667, -23.333333
4 / 180°	4	-149.99074, 1.666667, -23.333333
4 / 180°	5	-149.99074, 1.666667, 23.333333
4 / 180°	6	-101.662037, 1.666667, 23.333333
4 / 180°	7	-101.662037, 1.666667, 11.666667
4 / 180°	8	-53.333333, 1.666667, 11.666667

Blade / azimuth	Vertex	Local X, Y, Z mm
4 / 180°	9	-53.333333, -1.666667, -11.666667
4 / 180°	10	-101.662037, -1.666667, -11.666667
4 / 180°	11	-101.662037, -1.666667, -23.333333
4 / 180°	12	-149.99074, -1.666667, -23.333333
4 / 180°	13	-149.99074, -1.666667, 23.333333
4 / 180°	14	-101.662037, -1.666667, 23.333333
4 / 180°	15	-101.662037, -1.666667, 11.666667
4 / 180°	16	-53.333333, -1.666667, 11.666667
5 / 240°	1	-28.110042, -45.354688, -11.666667
5 / 240°	2	-52.274394, -87.208573, -11.666667
5 / 240°	3	-52.274394, -87.208573, -23.333333
5 / 240°	4	-76.438746, -129.062458, -23.333333
5 / 240°	5	-76.438746, -129.062458, 23.333333
5 / 240°	6	-52.274394, -87.208573, 23.333333
5 / 240°	7	-52.274394, -87.208573, 11.666667
5 / 240°	8	-28.110042, -45.354688, 11.666667
5 / 240°	9	-25.223291, -47.021355, -11.666667
5 / 240°	10	-49.387643, -88.87524, -11.666667
5 / 240°	11	-49.387643, -88.87524, -23.333333
5 / 240°	12	-73.551995, -130.729125, -23.333333
5 / 240°	13	-73.551995, -130.729125, 23.333333
5 / 240°	14	-49.387643, -88.87524, 23.333333
5 / 240°	15	-49.387643, -88.87524, 11.666667
5 / 240°	16	-25.223291, -47.021355, 11.666667
6 / 300°	1	25.223291, -47.021355, -11.666667
6 / 300°	2	49.387643, -88.87524, -11.666667
6 / 300°	3	49.387643, -88.87524, -23.333333
6 / 300°	4	73.551995, -130.729125, -23.333333
6 / 300°	5	73.551995, -130.729125, 23.333333
6 / 300°	6	49.387643, -88.87524, 23.333333
6 / 300°	7	49.387643, -88.87524, 11.666667
6 / 300°	8	25.223291, -47.021355, 11.666667
6 / 300°	9	28.110042, -45.354688, -11.666667
6 / 300°	10	52.274394, -87.208573, -11.666667
6 / 300°	11	52.274394, -87.208573, -23.333333

Blade / azimuth	Vertex	Local X, Y, Z mm
6 / 300°	12	76.438746, -129.062458, -23.333333
6 / 300°	13	76.438746, -129.062458, 23.333333
6 / 300°	14	52.274394, -87.208573, 23.333333
6 / 300°	15	52.274394, -87.208573, 11.666667
6 / 300°	16	28.110042, -45.354688, 11.666667

Stator geometry and area convention

Approved physical convention: $do = D \times \sqrt{\phi_s}$. Gross opening fraction = $(do/D)^2 = \phi_s$. Shaft-blocked net fraction = $(do^2 - ds^2)/D^2$. Preserve the original empirical Stage-3 ϕ_s ; neither physical fraction is fed back upstream.

Design feature	Saved value	Unit	Provenance / governing rule
Stator Free Area Ratio	0.4	—	Class A; Inherited Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Stator Thickness	3.75	mm	Class C; System-generated Class C. $\min(D/160, hc/20)$; geometric plate envelope.
Stator Opening Diameter	379.473319	mm	Class C; System-generated Class C. $D \sqrt{\phi_s}$; gross opening convention, no shaft compensation.
Stator Gross Open Area area	113097.335529	mm ²	Class C; System-generated Class C. $\pi do^2/4$.
Stator Net Open Area area	109606.677025	mm ²	Class C; System-generated Class C. $\pi(do^2 - ds^2)/4$; no upstream feedback.
Gross Free Area Ratio	0.4	—	Class C; System-generated Class C. $(do/D)^2 = \phi_s$.
Shaft Blocked Free Area Ratio	0.387654	—	Class C; System-generated Class C. $(do^2 - ds^2)/D^2$; separate physical report.
Calculated Free Area Ratio	0.4	—	Class C; System-generated Class C. Gross ratio, not legacy net-area convention.
Stator Radial Width	110.26334	mm	Class C; System-generated Class C. $(D - do)/2$.
Stator Construction	Annular single central opening	—	Class B; System-generated Class B. Garthe / Weber-Jupke reference family.

Shaft and shaft-support geometry

Design feature	Saved value	Unit	Provenance / governing rule
Shaft Diameter	66.666667	mm	Class C; System-generated Class C. (10/45) DR; reference-inspired scaling, not strength qualification.
Lower Shaft Support	450	mm	Class C; System-generated Class C. hB+LB/2.
Upper Shaft Support	8070	mm	Class C; System-generated Class C. z1+LT/2.
Support Arm Width	12	mm	Class C; System-generated Class C. 0.02D; tangential width, three arms at 30/150/270 degrees.
Support Arm Depth	24	mm	Class C; System-generated Class C. 0.04D; axial depth.
Support Housing Diameter	100	mm	Class C; System-generated Class C. 1.5ds; central envelope.
Support Housing Height	24	mm	Class C; System-generated Class C. 0.04D; central envelope.
Support Arm Radial Start	50	mm	Class C; System-generated Class C. 0.75ds; housing surface.
Support Arm Radial End	300	mm	Class C; System-generated Class C. D/2; shell interface.
Shaft Lower Termination	450	mm	Class C; System-generated Class C. Lower support centre.
Shaft Upper Termination	8670	mm	Class C; System-generated Class C. HV+D/4; drive engagement.
Shaft Length	8220	mm	Class C; System-generated Class C. Upper minus lower termination.
Shaft Area area	3490.658504	mm ²	Class C; System-generated Class C. pi ds ² /4.
Shaft Opening Clearance	156.403326	mm	Class C; System-generated Class C. (do-ds)/2.
Support Azimuths angle	30, 150, 270	deg	Class C; System-generated Class C. Approved three-arm support clocking.

Support	Centre Z mm	Housing OD mm	Housing height mm
1	450	100	24
2	8070	100	24

Support / arm / azimuth	Bottom / top Z mm	Ordered XY footprint mm
1 / 1 / 30°	438 / 462	46.30127, 19.803848; 262.807621, 144.803848; 256.807621, 155.196152; 40.30127, 30.196152
1 / 2 / 150°	438 / 462	-40.30127, 30.196152; -256.807621, 155.196152; -262.807621, 144.803848; -46.30127, 19.803848
1 / 3 / 270°	438 / 462	-6, -50; -6, -300; 6, -300; 6, -50
2 / 1 / 30°	8058 / 8082	46.30127, 19.803848; 262.807621, 144.803848; 256.807621, 155.196152; 40.30127, 30.196152

Support / arm / azimuth	Bottom / top Z mm	Ordered XY footprint mm
2 / 2 / 150°	8058 / 8082	-40.30127, 30.196152; -256.807621, 155.196152; -262.807621, 144.803848; -46.30127, 19.803848
2 / 3 / 270°	8058 / 8082	-6, -50; -6, -300; 6, -300; 6, -50

Skirt, access opening and seal/drive reserved envelopes

Design feature	Saved value	Unit	Provenance / governing rule
Support Height	300	mm	Class C; System-generated Class C. D/2; skirt below vessel datum.
Skirt Diameter	600	mm	Class C; System-generated Class C. D; centred cylindrical envelope.
Skirt Access Width	180	mm	Class C; System-generated Class C. 0.30D; access opening at azimuth zero.
Skirt Access Height	150	mm	Class C; System-generated Class C. 0.25D; access opening.
Skirt Access Elevation	-150	mm	Class C; System-generated Class C. -0.25D; centre elevation.
Seal Elevation	8520	mm	Class C; System-generated Class C. Top pole; coaxial with shaft.
Seal Diameter	133.333333	mm	Class C; System-generated Class C. 2ds; reserved housing, not product selection.
Seal Height	60	mm	Class C; System-generated Class C. 0.10D above top pole.
Drive Pedestal Diameter	300	mm	Class C; System-generated Class C. D/2; coaxial reserved envelope.
Drive Pedestal Height	150	mm	Class C; System-generated Class C. D/4 above top pole.
Drive Body Diameter	300	mm	Class C; System-generated Class C. D/2; reserved envelope.
Drive Body Height	450	mm	Class C; System-generated Class C. 3D/4 above pedestal.
Drive Height	600	mm	Class C; System-generated Class C. D including pedestal; not gearbox adequacy.
Skirt Access Azimuth angle	0	deg	Class C; System-generated Class C. Approved skirt access orientation.

Saved component envelope	Diameter mm	Bottom Z mm	Top Z mm
Skirt	600	-300	0
Drive-pedestal	300	8520	8670
Drive-body	300	8670	9120
Seal	133.333333	8520	8580
Shaft	66.666667	450	8670

Skirt access width mm	Height mm	Centre Z mm	Azimuth °
180	150	-150	0

Complete connection / nozzle schedule

All connections are preliminary geometric allowances, not standard DN selections or hydraulically qualified piping. Stub OD/projection are saved envelopes. No flange, reinforcement, weld or pipe schedule is implied.

Tag / service	Region / axis	Bore / OD mm	Projection mm	EL / radial offset mm	Azimuth °
P01 RRBO feed	bottom radial	60 / 72	60	510 / 300	0
P02 NMP-rich extract outlet	bottom radial	60 / 72	60	390 / 300	180
P03 NMP feed	top radial	60 / 72	60	8010 / 300	180
P04 RRBO-rich raffinate outlet	top radial	60 / 72	60	8130 / 300	0
A01 Low-shell flush	bottom radial	24 / 28.8	60	270 / 300	270
A02 High-shell equalization	top radial	24 / 28.8	60	8250 / 300	270
S01 Lower sample connection	bottom radial	15 / 18	60	630 / 300	90
S02 Upper sample connection	top radial	15 / 18	60	7890 / 300	90
I01 Lower instrument connection	bottom radial	15 / 18	60	330 / 300	45
I02 Upper instrument connection	top radial	15 / 18	60	7950 / 300	45
D01 Bottom-head drain	bottom-head down	24 / 28.8	60	0 / 0	N/A — axial
V01 Top-head crown-region vent	top-head up	24 / 28.8	60	8490 / 180	315

Tag / axis	Surface centre XYZ mm	External end XYZ mm	Head intersection Z min / max mm
P01 / radial	300, 0, 510	360, 0, 510	N/A — straight shell
P02 / radial	-300, 0, 390	-360, 0, 390	N/A — straight shell
P03 / radial	-300, 0, 8010	-360, 0, 8010	N/A — straight shell
P04 / radial	300, 0, 8130	360, 0, 8130	N/A — straight shell
A01 / radial	0, -300, 270	0, -360, 270	N/A — straight shell
A02 / radial	0, -300, 8250	0, -360, 8250	N/A — straight shell
S01 / radial	0, 300, 630	0, 360, 630	N/A — straight shell
S02 / radial	0, 300, 7890	0, 360, 7890	N/A — straight shell
I01 / radial	212.132034, 212.132034, 330	254.558441, 254.558441, 330	N/A — straight shell
I02 / radial	212.132034, 212.132034, 7950	254.558441, 254.558441, 7950	N/A — straight shell
D01 / down	0, 0, 0	0, 0, -60	0 / 0.1729
V01 / up	127.279221, -127.279221, 8490	127.279221, -127.279221, 8550	8484.24605 / 8495.076617

Tag	Saved connection rationale
Tag	Saved connection rationale
P01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P02	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P04	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
A01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A02	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S02	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
I01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
D01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A.
V01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: $r=0.30D$, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

Head penetration boundaries: every saved sample point follows in its stored order. Close each boundary loop; these samples describe the saved neck/head intersection, not a new machining allowance.

Connection	Boundary point	Global X, Y, Z mm
D01	1	14.4, 0, 0.1729
D01	2	14.33066, 1.411447, 0.1729
D01	3	14.123308, 2.809301, 0.1729
D01	4	13.779941, 4.180099, 0.1729
D01	5	13.303865, 5.510641, 0.1729
D01	6	12.699666, 6.788113, 0.1729
D01	7	11.973162, 8.000211, 0.1729
D01	8	11.131351, 9.135263, 0.1729
D01	9	10.182338, 10.182338, 0.1729
D01	10	9.135263, 11.131351, 0.1729
D01	11	8.000211, 11.973162, 0.1729
D01	12	6.788113, 12.699666, 0.1729
D01	13	5.510641, 13.303865, 0.1729

Connection	Boundary point	Global X, Y, Z mm
D01	14	4.180099, 13.779941, 0.1729
D01	15	2.809301, 14.123308, 0.1729
D01	16	1.411447, 14.33066, 0.1729
D01	17	0, 14.4, 0.1729
D01	18	-1.411447, 14.33066, 0.1729
D01	19	-2.809301, 14.123308, 0.1729
D01	20	-4.180099, 13.779941, 0.1729
D01	21	-5.510641, 13.303865, 0.1729
D01	22	-6.788113, 12.699666, 0.1729
D01	23	-8.000211, 11.973162, 0.1729
D01	24	-9.135263, 11.131351, 0.1729
D01	25	-10.182338, 10.182338, 0.1729
D01	26	-11.131351, 9.135263, 0.1729
D01	27	-11.973162, 8.000211, 0.1729
D01	28	-12.699666, 6.788113, 0.1729
D01	29	-13.303865, 5.510641, 0.1729
D01	30	-13.779941, 4.180099, 0.1729
D01	31	-14.123308, 2.809301, 0.1729
D01	32	-14.33066, 1.411447, 0.1729
D01	33	-14.4, 0, 0.1729
D01	34	-14.33066, -1.411447, 0.1729
D01	35	-14.123308, -2.809301, 0.1729
D01	36	-13.779941, -4.180099, 0.1729
D01	37	-13.303865, -5.510641, 0.1729
D01	38	-12.699666, -6.788113, 0.1729
D01	39	-11.973162, -8.000211, 0.1729
D01	40	-11.131351, -9.135263, 0.1729
D01	41	-10.182338, -10.182338, 0.1729
D01	42	-9.135263, -11.131351, 0.1729
D01	43	-8.000211, -11.973162, 0.1729
D01	44	-6.788113, -12.699666, 0.1729
D01	45	-5.510641, -13.303865, 0.1729
D01	46	-4.180099, -13.779941, 0.1729
D01	47	-2.809301, -14.123308, 0.1729
D01	48	-1.411447, -14.33066, 0.1729

Connection	Boundary point	Global X, Y, Z mm
D01	49	0, -14.4, 0.1729
D01	50	1.411447, -14.33066, 0.1729
D01	51	2.809301, -14.123308, 0.1729
D01	52	4.180099, -13.779941, 0.1729
D01	53	5.510641, -13.303865, 0.1729
D01	54	6.788113, -12.699666, 0.1729
D01	55	8.000211, -11.973162, 0.1729
D01	56	9.135263, -11.131351, 0.1729
D01	57	10.182338, -10.182338, 0.1729
D01	58	11.131351, -9.135263, 0.1729
D01	59	11.973162, -8.000211, 0.1729
D01	60	12.699666, -6.788113, 0.1729
D01	61	13.303865, -5.510641, 0.1729
D01	62	13.779941, -4.180099, 0.1729
D01	63	14.123308, -2.809301, 0.1729
D01	64	14.33066, -1.411447, 0.1729
V01	1	141.679221, -127.279221, 8485.895425
V01	2	141.609881, -125.867774, 8486.301274
V01	3	141.402529, -124.46992, 8486.739629
V01	4	141.059161, -123.099121, 8487.205929
V01	5	140.583086, -121.768579, 8487.695379
V01	6	139.978887, -120.491108, 8488.203008
V01	7	139.252383, -119.279009, 8488.723725
V01	8	138.410571, -118.143957, 8489.252379
V01	9	137.461558, -117.096883, 8489.783805
V01	10	136.414484, -116.14787, 8490.312885
V01	11	135.279432, -115.306058, 8490.834586
V01	12	134.067334, -114.579554, 8491.344011
V01	13	132.789862, -113.975355, 8491.836438
V01	14	131.45932, -113.49928, 8492.307359
V01	15	130.088521, -113.155913, 8492.752511
V01	16	128.690667, -112.948561, 8493.167909
V01	17	127.279221, -112.879221, 8493.54987
V01	18	125.867774, -112.948561, 8493.895042
V01	19	124.46992, -113.155913, 8494.200421

Connection	Boundary point	Global X, Y, Z mm
V01	20	123.099121, -113.49928, 8494.463368
V01	21	121.768579, -113.975355, 8494.681626
V01	22	120.491108, -114.579554, 8494.853333
V01	23	119.279009, -115.306058, 8494.977029
V01	24	118.143957, -116.14787, 8495.051667
V01	25	117.096883, -117.096883, 8495.076617
V01	26	116.14787, -118.143957, 8495.051667
V01	27	115.306058, -119.279009, 8494.977029
V01	28	114.579554, -120.491108, 8494.853333
V01	29	113.975355, -121.768579, 8494.681626
V01	30	113.49928, -123.099121, 8494.463368
V01	31	113.155913, -124.46992, 8494.200421
V01	32	112.948561, -125.867774, 8493.895042
V01	33	112.879221, -127.279221, 8493.54987
V01	34	112.948561, -128.690667, 8493.167909
V01	35	113.155913, -130.088521, 8492.752511
V01	36	113.49928, -131.45932, 8492.307359
V01	37	113.975355, -132.789862, 8491.836438
V01	38	114.579554, -134.067334, 8491.344011
V01	39	115.306058, -135.279432, 8490.834586
V01	40	116.14787, -136.414484, 8490.312885
V01	41	117.096883, -137.461558, 8489.783805
V01	42	118.143957, -138.410571, 8489.252379
V01	43	119.279009, -139.252383, 8488.723725
V01	44	120.491108, -139.978887, 8488.203008
V01	45	121.768579, -140.583086, 8487.695379
V01	46	123.099121, -141.059161, 8487.205929
V01	47	124.46992, -141.402529, 8486.739629
V01	48	125.867774, -141.609881, 8486.301274
V01	49	127.279221, -141.679221, 8485.895425
V01	50	128.690667, -141.609881, 8485.526354
V01	51	130.088521, -141.402529, 8485.197983
V01	52	131.45932, -141.059161, 8484.913837
V01	53	132.789862, -140.583086, 8484.67699
V01	54	134.067334, -139.978887, 8484.490023

Connection	Boundary point	Global X, Y, Z mm
V01	55	135.279432, -139.252383, 8484.354984
V01	56	136.414484, -138.410571, 8484.273359
V01	57	137.461558, -137.461558, 8484.24605
V01	58	138.410571, -136.414484, 8484.273359
V01	59	139.252383, -135.279432, 8484.354984
V01	60	139.978887, -134.067334, 8484.490023
V01	61	140.583086, -132.789862, 8484.67699
V01	62	141.059161, -131.45932, 8484.913837
V01	63	141.402529, -130.088521, 8485.197983
V01	64	141.609881, -128.690667, 8485.526354

Component quantities and complete elevation schedules

Every saved rotor, stator and compartment is listed. Stator plate thickness is centred on each boundary plane. These are not representative or truncated schedules.

Component	Quantity	Diameter mm	Thickness / axial size mm
R1 six-blade stepped rotor — Class C reference-inspired approximation	39	300	46.666667
Annular ring stator — single gross central opening	40	600	3.75
Continuous shaft envelope; mechanical qualification excluded	1	66.666667	N/A — solid shaft envelope
Three-arm shaft supports, azimuths 30/150/270 deg	2	600	24

Other modelled components	Quantity / definition
Head envelopes	2
Shaft-support assemblies	2
Support arms	6
Connections	12
Blades per rotor	6
Reserved component envelopes	Skirt, Drive-pedestal, Drive-body, Seal, Shaft
Skirt access opening	1

Compartment	Bottom Z mm	Top Z mm	Rotor centre Z mm	Saved stator Z mm
1	750	930	840	930
2	930	1110	1020	1110
3	1110	1290	1200	1290
4	1290	1470	1380	1470
5	1470	1650	1560	1650
6	1650	1830	1740	1830
7	1830	2010	1920	2010
8	2010	2190	2100	2190
9	2190	2370	2280	2370
10	2370	2550	2460	2550
11	2550	2730	2640	2730
12	2730	2910	2820	2910
13	2910	3090	3000	3090
14	3090	3270	3180	3270
15	3270	3450	3360	3450
16	3450	3630	3540	3630

Compartment	Bottom Z mm	Top Z mm	Rotor centre Z mm	Saved stator Z mm
17	3630	3810	3720	3810
18	3810	3990	3900	3990
19	3990	4170	4080	4170
20	4170	4350	4260	4350
21	4350	4530	4440	4530
22	4530	4710	4620	4710
23	4710	4890	4800	4890
24	4890	5070	4980	5070
25	5070	5250	5160	5250
26	5250	5430	5340	5430
27	5430	5610	5520	5610
28	5610	5790	5700	5790
29	5790	5970	5880	5970
30	5970	6150	6060	6150
31	6150	6330	6240	6330
32	6330	6510	6420	6510
33	6510	6690	6600	6690
34	6690	6870	6780	6870
35	6870	7050	6960	7050
36	7050	7230	7140	7230
37	7230	7410	7320	7410
38	7410	7590	7500	7590
39	7590	7770	7680	7770

R1 six-blade stepped rotor — Class C reference-inspired approximation — every saved elevation

Component ID	Ordinal	Centre / boundary Z mm
R	1	840
R	2	1020
R	3	1200
R	4	1380
R	5	1560
R	6	1740
R	7	1920
R	8	2100
R	9	2280

Component ID	Ordinal	Centre / boundary Z mm
R	10	2460
R	11	2640
R	12	2820
R	13	3000
R	14	3180
R	15	3360
R	16	3540
R	17	3720
R	18	3900
R	19	4080
R	20	4260
R	21	4440
R	22	4620
R	23	4800
R	24	4980
R	25	5160
R	26	5340
R	27	5520
R	28	5700
R	29	5880
R	30	6060
R	31	6240
R	32	6420
R	33	6600
R	34	6780
R	35	6960
R	36	7140
R	37	7320
R	38	7500
R	39	7680

Annular ring stator — single gross central opening — every saved elevation

Component ID	Ordinal	Centre / boundary Z mm
S	1	750
S	2	930

Component ID	Ordinal	Centre / boundary Z mm
S	3	1110
S	4	1290
S	5	1470
S	6	1650
S	7	1830
S	8	2010
S	9	2190
S	10	2370
S	11	2550
S	12	2730
S	13	2910
S	14	3090
S	15	3270
S	16	3450
S	17	3630
S	18	3810
S	19	3990
S	20	4170
S	21	4350
S	22	4530
S	23	4710
S	24	4890
S	25	5070
S	26	5250
S	27	5430
S	28	5610
S	29	5790
S	30	5970
S	31	6150
S	32	6330
S	33	6510
S	34	6690
S	35	6870
S	36	7050
S	37	7230

Component ID	Ordinal	Centre / boundary Z mm
S	38	7410
S	39	7590
S	40	7770

Continuous shaft envelope; mechanical qualification excluded — every saved elevation

Component ID	Ordinal	Centre / boundary Z mm
SH	1	450
SH	2	8670

Three-arm shaft supports, azimuths 30/150/270 deg — every saved elevation

Component ID	Ordinal	Centre / boundary Z mm
SUP	1	450
SUP	2	8070

Critical clearances

The saved clearances are geometric checks only; manufacturing tolerances, thermal movement and mechanical allowances are not introduced by this report.

Design feature	Saved value	Unit	Provenance / governing rule
Rotor Wall Clearance	150	mm	Class C; System-generated Class C. (D-DR)/2.
Shaft Opening Clearance	156.403326	mm	Class C; System-generated Class C. (do-ds)/2.
Stator Radial Width	110.26334	mm	Class C; System-generated Class C. (D-do)/2.
Rotor Lower Clearance	64.791667	mm	Class C; System-generated Class C. (hc-ts-HR)/2.
Rotor Upper Clearance	64.791667	mm	Class C; System-generated Class C. (hc-ts-HR)/2.

Validation results — complete saved check register

366 passed of 366. Status is from this saved revision, not a rerun against current upstream results.

Check	Result	Validation requirement
Authority	PASS	Frozen sources must be identified, current and compatible.
Column Diameter	PASS	columnDiameterM must be finite and positive.
Rotor Diameter	PASS	rotorDiameterM must be finite and positive.
Rotor Diameter Ratio	PASS	rotorDiameterRatio must be finite and positive.
Compartment Height	PASS	compartmentHeightM must be finite and positive.
Compartment Count	PASS	compartmentCount must be finite and positive.
Installed Active Height	PASS	installedActiveHeightM must be finite and positive.
Required Active Height	PASS	requiredActiveHeightM must be finite and positive.
Design Nt	PASS	designNt must be finite and positive.
Selected Rpm	PASS	selectedRpm must be finite and positive.
Rpm Min	PASS	rpmMin must be finite and positive.
Rpm Max	PASS	rpmMax must be finite and positive.
Stator Free Area Ratio	PASS	statorFreeAreaRatio must be finite and positive.
Count	PASS	Positive integer count; drawing capacity 1000 compartments.
Stack	PASS	Frozen count × pitch must equal frozen installed height; no repair.
Hets	PASS	Legacy HETS must be finite and positive.
Height	PASS	Frozen required height must agree with Nt × HETS and fit installed height.
Rotor-ratio	PASS	Frozen rotor diameter/ratio must agree and fit column.
Rpm	PASS	Frozen selected RPM must lie inside frozen window.
Phase	PASS	Frozen phase continuity is required.
Phi-domain	PASS	Empirical free-area input must be between zero and one.
Finite-column Diameter	PASS	columnDiameterM must be finite.
Finite-rotor Diameter	PASS	rotorDiameterM must be finite.
Finite-rotor Diameter Ratio	PASS	rotorDiameterRatio must be finite.
Finite-compartment Height	PASS	compartmentHeightM must be finite.
Finite-compartment Count	PASS	compartmentCount must be finite.
Finite-required Active Height	PASS	requiredActiveHeightM must be finite.
Finite-installed Active Height	PASS	installedActiveHeightM must be finite.
Finite-design Nt	PASS	designNt must be finite.
Finite-hets	PASS	hetsM must be finite.
Finite-stator Free Area Ratio	PASS	statorFreeAreaRatio must be finite.
Finite-selected Rpm	PASS	selectedRpm must be finite.
Finite-rpm Min	PASS	rpmMin must be finite.

Check	Result	Validation requirement
Finite-rpm Max	PASS	rpmMax must be finite.
Finite-shaft Diameter	PASS	shaftDiameterM must be finite.
Finite-stator Thickness	PASS	statorThicknessM must be finite.
Finite-blade Count	PASS	bladeCount must be finite.
Finite-blade Height	PASS	bladeHeightM must be finite.
Finite-hub Diameter	PASS	hubDiameterM must be finite.
Finite-hub Height	PASS	hubHeightM must be finite.
Finite-blade Thickness	PASS	bladeThicknessM must be finite.
Finite-blade Radial Length	PASS	bladeRadialLengthM must be finite.
Finite-blade Inner Segment Length	PASS	bladeInnerSegmentLengthM must be finite.
Finite-blade Outer Segment Length	PASS	bladeOuterSegmentLengthM must be finite.
Finite-blade Inner Segment Height	PASS	bladeInnerSegmentHeightM must be finite.
Finite-blade Outer Segment Height	PASS	bladeOuterSegmentHeightM must be finite.
Finite-rotor Offset	PASS	rotorOffsetM must be finite.
Finite-bottom Disengagement	PASS	bottomDisengagementM must be finite.
Finite-top Disengagement	PASS	topDisengagementM must be finite.
Finite-bottom Head Depth	PASS	bottomHeadDepthM must be finite.
Finite-top Head Depth	PASS	topHeadDepthM must be finite.
Finite-head Straight Flange	PASS	headStraightFlangeM must be finite.
Finite-support Height	PASS	supportHeightM must be finite.
Finite-skirt Diameter	PASS	skirtDiameterM must be finite.
Finite-skirt Access Width	PASS	skirtAccessWidthM must be finite.
Finite-skirt Access Height	PASS	skirtAccessHeightM must be finite.
Finite-skirt Access Elevation	PASS	skirtAccessElevationM must be finite.
Finite-lower Shaft Support	PASS	lowerShaftSupportM must be finite.
Finite-upper Shaft Support	PASS	upperShaftSupportM must be finite.
Finite-support Arm Width	PASS	supportArmWidthM must be finite.
Finite-support Arm Depth	PASS	supportArmDepthM must be finite.
Finite-support Housing Diameter	PASS	supportHousingDiameterM must be finite.
Finite-support Housing Height	PASS	supportHousingHeightM must be finite.
Finite-support Arm Radial Start	PASS	supportArmRadialStartM must be finite.
Finite-support Arm Radial End	PASS	supportArmRadialEndM must be finite.
Finite-shaft Lower Termination	PASS	shaftLowerTerminationM must be finite.

Check	Result	Validation requirement
Finite-shaft Upper Termination	PASS	shaftUpperTerminationM must be finite.
Finite-shaft Length	PASS	shaftLengthM must be finite.
Finite-seal Elevation	PASS	sealElevationM must be finite.
Finite-seal Diameter	PASS	sealDiameterM must be finite.
Finite-seal Height	PASS	sealHeightM must be finite.
Finite-drive Pedestal Diameter	PASS	drivePedestalDiameterM must be finite.
Finite-drive Pedestal Height	PASS	drivePedestalHeightM must be finite.
Finite-drive Body Diameter	PASS	driveBodyDiameterM must be finite.
Finite-drive Body Height	PASS	driveBodyHeightM must be finite.
Finite-drive Height	PASS	driveHeightM must be finite.
Finite-stator Opening Diameter	PASS	statorOpeningDiameterM must be finite.
Finite-column Area area	PASS	columnAreaM2 must be finite.
Finite-shaft Area area	PASS	shaftAreaM2 must be finite.
Finite-stator Gross Open Area area	PASS	statorGrossOpenAreaM2 must be finite.
Finite-stator Net Open Area area	PASS	statorNetOpenAreaM2 must be finite.
Finite-gross Free Area Ratio	PASS	grossFreeAreaRatio must be finite.
Finite-shaft Blocked Free Area Ratio	PASS	shaftBlockedFreeAreaRatio must be finite.
Finite-calculated Free Area Ratio	PASS	calculatedFreeAreaRatio must be finite.
Finite-rotor Wall Clearance	PASS	rotorWallClearanceM must be finite.
Finite-shaft Opening Clearance	PASS	shaftOpeningClearanceM must be finite.
Finite-stator Radial Width	PASS	statorRadialWidthM must be finite.
Finite-rotor Axial Envelope	PASS	rotorAxialEnvelopeM must be finite.
Finite-rotor Lower Clearance	PASS	rotorLowerClearanceM must be finite.
Finite-rotor Upper Clearance	PASS	rotorUpperClearanceM must be finite.
Finite-active Start	PASS	activeStartM must be finite.
Finite-active End	PASS	activeEndM must be finite.
Finite-top Tangent	PASS	topTangentM must be finite.
Finite-vessel Height	PASS	vesselHeightM must be finite.
Finite-overall Height	PASS	overallHeightM must be finite.
Finite-skirt Access Azimuth angle	PASS	skirtAccessAzimuthDeg must be finite.
Finite-P01-elevation	PASS	P01-elevationM must be finite.
Finite-P01-bore	PASS	P01-boreM must be finite.
Finite-P01-azimuth angle	PASS	P01-azimuthDeg must be finite.
Finite-P01-radial Offset	PASS	P01-radialOffsetM must be finite.

Check	Result	Validation requirement
Finite-P01-outside Diameter	PASS	P01-outsideDiameterM must be finite.
Finite-P01-projection	PASS	P01-projectionM must be finite.
Finite-P02-elevation	PASS	P02-elevationM must be finite.
Finite-P02-bore	PASS	P02-boreM must be finite.
Finite-P02-azimuth angle	PASS	P02-azimuthDeg must be finite.
Finite-P02-radial Offset	PASS	P02-radialOffsetM must be finite.
Finite-P02-outside Diameter	PASS	P02-outsideDiameterM must be finite.
Finite-P02-projection	PASS	P02-projectionM must be finite.
Finite-P03-elevation	PASS	P03-elevationM must be finite.
Finite-P03-bore	PASS	P03-boreM must be finite.
Finite-P03-azimuth angle	PASS	P03-azimuthDeg must be finite.
Finite-P03-radial Offset	PASS	P03-radialOffsetM must be finite.
Finite-P03-outside Diameter	PASS	P03-outsideDiameterM must be finite.
Finite-P03-projection	PASS	P03-projectionM must be finite.
Finite-P04-elevation	PASS	P04-elevationM must be finite.
Finite-P04-bore	PASS	P04-boreM must be finite.
Finite-P04-azimuth angle	PASS	P04-azimuthDeg must be finite.
Finite-P04-radial Offset	PASS	P04-radialOffsetM must be finite.
Finite-P04-outside Diameter	PASS	P04-outsideDiameterM must be finite.
Finite-P04-projection	PASS	P04-projectionM must be finite.
Finite-A01-elevation	PASS	A01-elevationM must be finite.
Finite-A01-bore	PASS	A01-boreM must be finite.
Finite-A01-azimuth angle	PASS	A01-azimuthDeg must be finite.
Finite-A01-radial Offset	PASS	A01-radialOffsetM must be finite.
Finite-A01-outside Diameter	PASS	A01-outsideDiameterM must be finite.
Finite-A01-projection	PASS	A01-projectionM must be finite.
Finite-A02-elevation	PASS	A02-elevationM must be finite.
Finite-A02-bore	PASS	A02-boreM must be finite.
Finite-A02-azimuth angle	PASS	A02-azimuthDeg must be finite.
Finite-A02-radial Offset	PASS	A02-radialOffsetM must be finite.
Finite-A02-outside Diameter	PASS	A02-outsideDiameterM must be finite.
Finite-A02-projection	PASS	A02-projectionM must be finite.
Finite-S01-elevation	PASS	S01-elevationM must be finite.
Finite-S01-bore	PASS	S01-boreM must be finite.
Finite-S01-azimuth angle	PASS	S01-azimuthDeg must be finite.

Check	Result	Validation requirement
Finite-S01-radial Offset	PASS	S01-radialOffsetM must be finite.
Finite-S01-outside Diameter	PASS	S01-outsideDiameterM must be finite.
Finite-S01-projection	PASS	S01-projectionM must be finite.
Finite-S02-elevation	PASS	S02-elevationM must be finite.
Finite-S02-bore	PASS	S02-boreM must be finite.
Finite-S02-azimuth angle	PASS	S02-azimuthDeg must be finite.
Finite-S02-radial Offset	PASS	S02-radialOffsetM must be finite.
Finite-S02-outside Diameter	PASS	S02-outsideDiameterM must be finite.
Finite-S02-projection	PASS	S02-projectionM must be finite.
Finite-I01-elevation	PASS	I01-elevationM must be finite.
Finite-I01-bore	PASS	I01-boreM must be finite.
Finite-I01-azimuth angle	PASS	I01-azimuthDeg must be finite.
Finite-I01-radial Offset	PASS	I01-radialOffsetM must be finite.
Finite-I01-outside Diameter	PASS	I01-outsideDiameterM must be finite.
Finite-I01-projection	PASS	I01-projectionM must be finite.
Finite-I02-elevation	PASS	I02-elevationM must be finite.
Finite-I02-bore	PASS	I02-boreM must be finite.
Finite-I02-azimuth angle	PASS	I02-azimuthDeg must be finite.
Finite-I02-radial Offset	PASS	I02-radialOffsetM must be finite.
Finite-I02-outside Diameter	PASS	I02-outsideDiameterM must be finite.
Finite-I02-projection	PASS	I02-projectionM must be finite.
Finite-D01-elevation	PASS	D01-elevationM must be finite.
Finite-D01-bore	PASS	D01-boreM must be finite.
Finite-D01-radial Offset	PASS	D01-radialOffsetM must be finite.
Finite-D01-outside Diameter	PASS	D01-outsideDiameterM must be finite.
Finite-D01-projection	PASS	D01-projectionM must be finite.
Finite-D01-surface Edge Elevation Min	PASS	D01-surfaceEdgeElevationMinM must be finite.
Finite-D01-surface Edge Elevation Max	PASS	D01-surfaceEdgeElevationMaxM must be finite.
Head-V01-radial-fit	PASS	Entire neck cylinder fits ellipsoidal head.
Finite-V01-elevation	PASS	V01-elevationM must be finite.
Finite-V01-bore	PASS	V01-boreM must be finite.
Finite-V01-azimuth angle	PASS	V01-azimuthDeg must be finite.
Finite-V01-radial Offset	PASS	V01-radialOffsetM must be finite.
Finite-V01-outside Diameter	PASS	V01-outsideDiameterM must be finite.

Check	Result	Validation requirement
Finite-V01-projection	PASS	V01-projectionM must be finite.
Finite-V01-surface Edge Elevation Min	PASS	V01-surfaceEdgeElevationMinM must be finite.
Finite-V01-surface Edge Elevation Max	PASS	V01-surfaceEdgeElevationMaxM must be finite.
Finite-compartment1-bottom	PASS	compartment1-bottomM must be finite.
Finite-compartment1-top	PASS	compartment1-topM must be finite.
Finite-compartment1-rotor	PASS	compartment1-rotorM must be finite.
Finite-compartment2-bottom	PASS	compartment2-bottomM must be finite.
Finite-compartment2-top	PASS	compartment2-topM must be finite.
Finite-compartment2-rotor	PASS	compartment2-rotorM must be finite.
Finite-compartment3-bottom	PASS	compartment3-bottomM must be finite.
Finite-compartment3-top	PASS	compartment3-topM must be finite.
Finite-compartment3-rotor	PASS	compartment3-rotorM must be finite.
Finite-compartment4-bottom	PASS	compartment4-bottomM must be finite.
Finite-compartment4-top	PASS	compartment4-topM must be finite.
Finite-compartment4-rotor	PASS	compartment4-rotorM must be finite.
Finite-compartment5-bottom	PASS	compartment5-bottomM must be finite.
Finite-compartment5-top	PASS	compartment5-topM must be finite.
Finite-compartment5-rotor	PASS	compartment5-rotorM must be finite.
Finite-compartment6-bottom	PASS	compartment6-bottomM must be finite.
Finite-compartment6-top	PASS	compartment6-topM must be finite.
Finite-compartment6-rotor	PASS	compartment6-rotorM must be finite.
Finite-compartment7-bottom	PASS	compartment7-bottomM must be finite.
Finite-compartment7-top	PASS	compartment7-topM must be finite.
Finite-compartment7-rotor	PASS	compartment7-rotorM must be finite.
Finite-compartment8-bottom	PASS	compartment8-bottomM must be finite.
Finite-compartment8-top	PASS	compartment8-topM must be finite.
Finite-compartment8-rotor	PASS	compartment8-rotorM must be finite.
Finite-compartment9-bottom	PASS	compartment9-bottomM must be finite.
Finite-compartment9-top	PASS	compartment9-topM must be finite.
Finite-compartment9-rotor	PASS	compartment9-rotorM must be finite.
Finite-compartment10-bottom	PASS	compartment10-bottomM must be finite.
Finite-compartment10-top	PASS	compartment10-topM must be finite.
Finite-compartment10-rotor	PASS	compartment10-rotorM must be finite.
Finite-compartment11-bottom	PASS	compartment11-bottomM must be finite.

Check	Result	Validation requirement
Finite-compartment11-top	PASS	compartment11-topM must be finite.
Finite-compartment11-rotor	PASS	compartment11-rotorM must be finite.
Finite-compartment12-bottom	PASS	compartment12-bottomM must be finite.
Finite-compartment12-top	PASS	compartment12-topM must be finite.
Finite-compartment12-rotor	PASS	compartment12-rotorM must be finite.
Finite-compartment13-bottom	PASS	compartment13-bottomM must be finite.
Finite-compartment13-top	PASS	compartment13-topM must be finite.
Finite-compartment13-rotor	PASS	compartment13-rotorM must be finite.
Finite-compartment14-bottom	PASS	compartment14-bottomM must be finite.
Finite-compartment14-top	PASS	compartment14-topM must be finite.
Finite-compartment14-rotor	PASS	compartment14-rotorM must be finite.
Finite-compartment15-bottom	PASS	compartment15-bottomM must be finite.
Finite-compartment15-top	PASS	compartment15-topM must be finite.
Finite-compartment15-rotor	PASS	compartment15-rotorM must be finite.
Finite-compartment16-bottom	PASS	compartment16-bottomM must be finite.
Finite-compartment16-top	PASS	compartment16-topM must be finite.
Finite-compartment16-rotor	PASS	compartment16-rotorM must be finite.
Finite-compartment17-bottom	PASS	compartment17-bottomM must be finite.
Finite-compartment17-top	PASS	compartment17-topM must be finite.
Finite-compartment17-rotor	PASS	compartment17-rotorM must be finite.
Finite-compartment18-bottom	PASS	compartment18-bottomM must be finite.
Finite-compartment18-top	PASS	compartment18-topM must be finite.
Finite-compartment18-rotor	PASS	compartment18-rotorM must be finite.
Finite-compartment19-bottom	PASS	compartment19-bottomM must be finite.
Finite-compartment19-top	PASS	compartment19-topM must be finite.
Finite-compartment19-rotor	PASS	compartment19-rotorM must be finite.
Finite-compartment20-bottom	PASS	compartment20-bottomM must be finite.
Finite-compartment20-top	PASS	compartment20-topM must be finite.
Finite-compartment20-rotor	PASS	compartment20-rotorM must be finite.
Finite-compartment21-bottom	PASS	compartment21-bottomM must be finite.
Finite-compartment21-top	PASS	compartment21-topM must be finite.
Finite-compartment21-rotor	PASS	compartment21-rotorM must be finite.
Finite-compartment22-bottom	PASS	compartment22-bottomM must be finite.
Finite-compartment22-top	PASS	compartment22-topM must be finite.
Finite-compartment22-rotor	PASS	compartment22-rotorM must be finite.

Check	Result	Validation requirement
Finite-compartment23-bottom	PASS	compartment23-bottomM must be finite.
Finite-compartment23-top	PASS	compartment23-topM must be finite.
Finite-compartment23-rotor	PASS	compartment23-rotorM must be finite.
Finite-compartment24-bottom	PASS	compartment24-bottomM must be finite.
Finite-compartment24-top	PASS	compartment24-topM must be finite.
Finite-compartment24-rotor	PASS	compartment24-rotorM must be finite.
Finite-compartment25-bottom	PASS	compartment25-bottomM must be finite.
Finite-compartment25-top	PASS	compartment25-topM must be finite.
Finite-compartment25-rotor	PASS	compartment25-rotorM must be finite.
Finite-compartment26-bottom	PASS	compartment26-bottomM must be finite.
Finite-compartment26-top	PASS	compartment26-topM must be finite.
Finite-compartment26-rotor	PASS	compartment26-rotorM must be finite.
Finite-compartment27-bottom	PASS	compartment27-bottomM must be finite.
Finite-compartment27-top	PASS	compartment27-topM must be finite.
Finite-compartment27-rotor	PASS	compartment27-rotorM must be finite.
Finite-compartment28-bottom	PASS	compartment28-bottomM must be finite.
Finite-compartment28-top	PASS	compartment28-topM must be finite.
Finite-compartment28-rotor	PASS	compartment28-rotorM must be finite.
Finite-compartment29-bottom	PASS	compartment29-bottomM must be finite.
Finite-compartment29-top	PASS	compartment29-topM must be finite.
Finite-compartment29-rotor	PASS	compartment29-rotorM must be finite.
Finite-compartment30-bottom	PASS	compartment30-bottomM must be finite.
Finite-compartment30-top	PASS	compartment30-topM must be finite.
Finite-compartment30-rotor	PASS	compartment30-rotorM must be finite.
Finite-compartment31-bottom	PASS	compartment31-bottomM must be finite.
Finite-compartment31-top	PASS	compartment31-topM must be finite.
Finite-compartment31-rotor	PASS	compartment31-rotorM must be finite.
Finite-compartment32-bottom	PASS	compartment32-bottomM must be finite.
Finite-compartment32-top	PASS	compartment32-topM must be finite.
Finite-compartment32-rotor	PASS	compartment32-rotorM must be finite.
Finite-compartment33-bottom	PASS	compartment33-bottomM must be finite.
Finite-compartment33-top	PASS	compartment33-topM must be finite.
Finite-compartment33-rotor	PASS	compartment33-rotorM must be finite.
Finite-compartment34-bottom	PASS	compartment34-bottomM must be finite.
Finite-compartment34-top	PASS	compartment34-topM must be finite.

Check	Result	Validation requirement
Finite-compartment34-rotor	PASS	compartment34-rotorM must be finite.
Finite-compartment35-bottom	PASS	compartment35-bottomM must be finite.
Finite-compartment35-top	PASS	compartment35-topM must be finite.
Finite-compartment35-rotor	PASS	compartment35-rotorM must be finite.
Finite-compartment36-bottom	PASS	compartment36-bottomM must be finite.
Finite-compartment36-top	PASS	compartment36-topM must be finite.
Finite-compartment36-rotor	PASS	compartment36-rotorM must be finite.
Finite-compartment37-bottom	PASS	compartment37-bottomM must be finite.
Finite-compartment37-top	PASS	compartment37-topM must be finite.
Finite-compartment37-rotor	PASS	compartment37-rotorM must be finite.
Finite-compartment38-bottom	PASS	compartment38-bottomM must be finite.
Finite-compartment38-top	PASS	compartment38-topM must be finite.
Finite-compartment38-rotor	PASS	compartment38-rotorM must be finite.
Finite-compartment39-bottom	PASS	compartment39-bottomM must be finite.
Finite-compartment39-top	PASS	compartment39-topM must be finite.
Finite-compartment39-rotor	PASS	compartment39-rotorM must be finite.
Shaft-opening	PASS	Shaft must fit gross opening and opening must fit column.
Hub-fit	PASS	Shaft < hub < rotor and positive radial reach.
Blade-spacing	PASS	Six blade roots do not overlap.
Swept-diameter	PASS	Actual outer-corner sweep equals frozen diameter.
Axial-clearance	PASS	Rotor/plates clear; terminal half-plates fit end zones.
Gross-area	PASS	Gross phi equals frozen empirical input; net reported separately.
Supports	PASS	Full support axial envelopes bracket stack inside straight shell.
Drive-seal-fit	PASS	Seal reserved envelope fits drive pedestal; shaft deliberately engages seal, supports and drive.
Skirt-drain	PASS	Drain neck fits inside skirt and remains above ground.
P01-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
P01-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
P02-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
P02-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
P03-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
P03-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
P04-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
P04-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.

Check	Result	Validation requirement
A01-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
A01-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
A02-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
A02-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
S01-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
S01-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
S02-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
S02-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
I01-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
I01-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
I02-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
I02-internals-envelope	PASS	Full neck envelope is axially disjoint from every stator and all support arms/housing.
P01-P02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-P03-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-P04-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-A01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-P03-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-P04-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-A01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-P04-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-A01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).

Check	Result	Validation requirement
P03-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-A01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A02-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A02-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A02-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A02-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S01-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S01-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S01-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S02-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S02-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
I01-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
Vent-drive-seal-envelope	PASS	Vertical vent cylinder clears full drive/pedestal/seal radial envelopes.
Head-connections-internals	PASS	Actual ellipsoid/neck intersection envelopes clear internal supports and shaft lower termination.
Head-shell-connections	PASS	Head connections and straight-shell stubs are axially disjoint.

R1 rules, source hashes and retained lineage

These identifiers belong to the saved revision. Missing legacy metadata is explicitly reported, never substituted with current sources.

Stage 3 saved reference	Value
Id	3
Immutable Hash	bb
Result / Calculation Hash	dd

Stage 4 saved reference	Value
Id	4
Lineage Hash	cc

Frozen rules manifest	Value
Id	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1
Revision	1
Units	m, m², rpm, degrees; vessel bottom datum
Authority	Frozen Stage3/4 values unchanged; no recalculation or feedback
Evidence / A	Frozen Stage3/4 authority
Evidence / B	Documented reference construction family
Evidence / C	Approved R1 engineering geometry
Rules / Opening	do=D*sqrt(phi_s); gross=(do/D)^2; net=(do^2-ds^2)/D^2; no feedback
Rules / Shaft	ds=(10/45)DR; lower=hB+LB/2; upper=HV+D/4
Rules / Stator	ts=min(D/160,hc/20); N+1 shared boundaries including terminal rings
Rules / Rotor	six blades; HR=min((7/45)DR,0.60(hc-ts)); dh=1.60ds; Hh=.80HR; tb=min(DR/90,HR/10,dh/12); Lb=(sqrt(DR²-tb²)-dh)/2
Rules / Step	inner Lb/2 height HR/2; outer Lb/2 height HR; symmetric axial profile; azimuths 0,60,120,180,240,300
Rules / Stack	rotor centres z0+(i+.5)hc; stators z0+i hc; z0=1.25D; HA=N hc
Rules / Endzones	LB=LT=D; no residence-time claim
Rules / Heads	2:1 ellipse; radial semiaxis D/2; axial semiaxis D/4; straight flange zero
Rules / Supports	skirt D diameter D/2 height; access .30D by .25D at z=-.25D, azimuth0; supports at hB+LB/2,z1+LT/2; arms azimuth30,150,270 width.02D depth.04D span .75ds to D/2; housingOD1.5ds height.04D
Rules / Drive	sealOD2ds height.10D at HV; pedestalOD D/2 heightD/4; bodyOD D/2 height3D/4; totalheightD
Rules / Routing	NMP down/RRBO up; frozen phase continuity unchanged
Rules / Connections	process bore D/10; drain/vent/flush/equalization D/25; sample/instrument D/40; OD1.20bore; projectionD/10
Rules / Schedule	P01 lower .60 0; P02 lower .40 180; P03 upper .40 180; P04 upper .60 0; A01 lower .20 270; A02 upper .80 270; S01 lower .80 90; S02 upper .20 90; I01 lower .30 45; I02 upper .30 45; D01 bottom pole down; V01 top head r=.30D azimuth315 up

Frozen rules manifest	Value
Rules / Intersections	Exact ellipsoid/cylinder boundary; full OD neck and projected envelope clearances; intended shaft/hub/support/drive interfaces excluded from clashes
Exclusions	Fabrication, structural and shaft-dynamic qualification, hydraulic nozzle sizing and disengagement performance; no user-entered dimensions

Additional saved design selections and evidence

Supplementary saved parameters not repeated in the component sections. All dimensions retain their saved meaning.

Design feature	Saved value	Unit	Provenance / governing rule
Flow Arrangement	nmp-down-rrbo-up	—	Class C; System-generated Class C. Explicit application layout; continuity and upstream hydraulics unchanged.
Design Envelope	hc/D 0.20–0.30; DR/ D 0.33–0.50; phi_s 0.20–0.40	—	Class B; System-generated Class B. Perry 8e p15-83 → Pratt-Stevens; advisory, never clamp inherited geometry.
P01-elevation	510	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-bore	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-azimuth angle	0	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-outside Diameter	72	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P02-elevation	390	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-bore	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-azimuth angle	180	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-outside Diameter	72	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
P02-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-elevation	8010	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-bore	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-azimuth angle	180	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-outside Diameter	72	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P04-elevation	8130	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-bore	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-azimuth angle	0	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-outside Diameter	72	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
A01-elevation	270	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
A01-bore	24	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A01-azimuth angle	270	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A01-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A01-outside Diameter	28.8	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A01-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A02-elevation	8250	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-bore	24	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-azimuth angle	270	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-outside Diameter	28.8	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-elevation	630	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-bore	15	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-azimuth angle	90	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
S01-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-outside Diameter	18	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S02-elevation	7890	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-bore	15	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-azimuth angle	90	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-outside Diameter	18	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
I01-elevation	330	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-bore	15	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-azimuth angle	45	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-outside Diameter	18	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
I01-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-elevation	7950	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-bore	15	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-azimuth angle	45	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-radial Offset	300	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-outside Diameter	18	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
D01-elevation	0	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A.
D01-bore	24	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A.
D01-radial Offset	0	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A.
D01-outside Diameter	28.8	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A.
D01-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A.
D01-surface Edge Elevation Min	0	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A.
D01-surface Edge Elevation Max	0.1729	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: bottom-pole axial drain; azimuth N/A.
V01-elevation	8490	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: $r=0.30D$, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

Design feature	Saved value	Unit	Provenance / governing rule
V01-bore	24	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-azimuth angle	315	deg	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-radial Offset	180	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-outside Diameter	28.8	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-projection	60	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-surface Edge Elevation Min	8484.24605	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-surface Edge Elevation Max	8495.076617	mm	Class C; System-generated Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R1; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

Limitations and CAD handoff acceptance

Geometrically complete pre-pilot layout only; not fabrication or mechanical qualification.

- Class A: approved frozen Stage-3/4; no hydraulic/sizing recalculation.
- Class B: Perry 8e p15-83 / Pratt-Stevens design envelope; Garthe Table A.1 and Weber-Jupke 2020 DOI 10.1002/aic.16286 construction family.
- Class C: approved R1 proportions and stepped approximation, not exact reference reproduction or experimental validation.
- End zones are packaging allocations, not separation-performance sizing. Connection bores are geometric allowances, not hydraulic or DN sizing.
- NMP-down/RRBO-up is an explicit layout selection, not inferred from continuity. Vent is crown-region, not absolute-high-point qualified.
- N+1 plates follows selected shared-boundary and terminal-ring topology, not a universal source count.
- Pressure-vessel shell/head thickness and code design
- Shaft strength, deflection, critical speed
- Bearing/seal ratings, drive power/torque and support structural qualification
- Welds, keys, fasteners, flanges, reinforcement and external pipe routing
- Separation residence-time, connection hydraulic capacity and operational venting qualification

Not specified by R1 — outside scope: pressure-rated wall/head material thicknesses; materials and corrosion allowance; welding, tolerances, fits and fasteners; bearing/seal product selection; shaft strength, deflection and critical speed; drive torque/power adequacy; structural loads; nozzle flanges/reinforcement and piping qualification. Do not invent these details or issue fabrication drawings from this report.

No unresolved geometry items recorded in this saved revision. Explicit scope exclusions above are not fabrication-ready details.

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION